

Stage 1 Workflow Summary: From Raw Data to Reference-Site Threshold Selection

1 Overview

This document summarises the workflow implemented in `src/run_stage_threshold.py`. The script runs two connected analyses in sequence:

1. Stage 1 contamination scoring from sediment chemistry (SumRel and MaxRel).
2. Reference-threshold sensitivity analysis using RDA over a grid of threshold proportions.

Together, these steps produce:

- site-level contamination scores,
- ecological evidence for score choice,
- a data-supported reference-site cutoff (25%).

The script reads `data/processed/complete_env_taxa_chemical_Feb.3.xlsx` and writes outputs to:

- `results/01_pollution_assessment/` (Stage 1 outputs),
- `results/ref_threshold_sensitivity/` (threshold sweep outputs).

2 Step 1: PCA-Based Contamination Scoring (Stage 1)

1. **Data ingestion.** The pipeline reads the three-level MultiIndex workbook (`complete_env_taxa_chemical_Feb.3`). From the (`chemical`, `raw`) block it extracts the 16 contaminant columns: Co, Al, Ni, Mn, Fe, Cr, Cu, Hg, Pb, Zn, total PCB, Cd, OCS, *p,p'*-DDE, As, Ca. Variables with negligible variance ($\sigma < 10^{-10}$) are dropped.
2. **Log-transformation.** Each contaminant concentration is transformed element-wise:

$$x' = \log_2(1 + x).$$

3. **Z-score normalisation.** Every log-transformed column is standardised to zero mean and unit variance:

$$z_{ij} = \frac{x'_{ij} - \bar{x}'_j}{\sigma_{x'_j}}.$$

This ensures PCA is not dominated by variables with larger absolute magnitudes after log-transformation.

4. **PCA fitting.** Scikit-learn PCA is applied to the z -scored matrix. The first $K = 5$ components are retained (`n_components=5`; cumulative explained variance $\gtrsim 83\%$). Scaled loadings are computed as $L_{jk} = v_{jk}\sqrt{\lambda_k}$.
5. **Sign orientation.** Each PC is inspected: if the variable with the largest absolute loading has a *negative* sign, both the loadings and site scores for that PC are flipped, so that higher scores always correspond to higher contamination.
6. **PC-score standardisation.** The oriented raw PC scores are min-max rescaled per component:

$$S'_{ik} = \frac{S_{ik} - \min_i S_k}{\max_i S_k - \min_i S_k}, \quad S'_{ik} \in [0, 1].$$

7. **Composite scoring rules.** From the rescaled scores S'_{ik} ($k = 1, \dots, 5$) two site-level indices are computed:

- **SumRel** — cumulative pressure: $\text{SumRel}_i = \sum_{k=1}^K S'_{ik}$.
- **MaxRel** — worst-axis pressure: $\text{MaxRel}_i = \max_{k=1, \dots, K} S'_{ik}$.

Outputs saved to `results/01_pollution_assessment/`:

- **Tables:** `pc_loadings` (scaled loadings + variance info), `site_scores` (standardised PC scores), `SumRel_site_rankings`, `MaxRel_site_rankings`.
- **Artifacts:** `SumRel_01_updated_data.xlsx`, `MaxRel_01_updated_data.xlsx` (full MultiIndex data augmented with PC scores and composite score).
- **Figures:** variance-explained plot, ridge-loading plot, corridor-bifurcation maps for SumRel and MaxRel (using shapefiles from `data/maps/`).

3 Step 2: Threshold Sensitivity Sweep

For each contamination score (SumRel and MaxRel), the pipeline evaluates how RDA performance changes as the reference-subset cutoff varies.

1. **Prepare response and predictor matrices.** From the same workbook, the `(taxa, raw)` block supplies the taxa abundance matrix $\mathbf{Y}$ and the `(environmental, raw)` block supplies five predictor variables $\mathbf{X}$: Measured Depth, Bottom DO, Temperature, MPS (Phi), and LOI (%).
2. **Define the threshold grid.** $q \in \{0.05, 0.07, \dots, 0.99\}$ (5 % to 99 %, step 2 %).
3. **At each threshold q (loop):**
 - (a) **Select reference sites.** Compute $n = \lfloor N \times q \rfloor$; take the n sites with the smallest contamination score.
 - (b) **Subset $\mathbf{X}$ and $\mathbf{Y}$** to the reference sites. Drop any rows with missing values.
 - (c) **Transform predictors.** Z-score each environmental column within the reference subset ($\mu = 0$, $\sigma = 1$; `standardize_env=True`). No log-transform is applied to environmental variables (`log_transform_env=False`).

- (d) **Transform taxa.** With `taxa_transform="octave"` the raw (already octave-transformed) abundances are used as-is.
- (e) **Fit RDA.** Both $\mathbf{X}$ and $\mathbf{Y}$ are mean-centred. Fitted values $\hat{\mathbf{Y}} = \mathbf{X}_c(\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{Y}_c$, then eigendecomposition of $\text{Cov}(\hat{\mathbf{Y}})$ yields constrained axes. Inertia partitioning gives R^2 and

$$R_{\text{adj}}^2 = 1 - (1 - R^2) \frac{n - 1}{df_{\text{res}}}.$$

- (f) **Global permutation test.** Observed pseudo- $F = (SS_{\text{con}}/df_{\text{model}}) / (SS_{\text{res}}/df_{\text{res}})$. Rows of $\mathbf{Y}_c$ are permuted 999 times; $p\text{-value} = (1 + \#\{F_\pi \geq F_{\text{obs}}\}) / (n_{\text{perm}} + 1)$.
 - (g) **Record metrics.** R^2 , R_{adj}^2 , pseudo- F , $p\text{-value}$, constrained inertia, total inertia, RDA1 eigenvalue, n sites.
4. **Aggregate.** Metrics across all thresholds are collected into a per-score table and a combined comparison table.
 5. **Between-score comparison plots.** Threshold curves for SumRel and MaxRel are overlaid for R_{adj}^2 and pseudo- F , enabling direct comparison.

4 Step 3: Full RDA at the Selected Threshold and Threshold Selection

1. **Full RDA fit at $q = 0.25$.** For each score the bottom-25 % reference sites are selected. Environmental variables are z-scored and both matrices are mean-centred. In addition to the global permutation test, two further tests are run:
 - **Per-axis test:** each constrained axis (RDA1, RDA2, ...) is tested individually for significance.
 - **Per-term (Freedman–Lane) test:** each environmental variable's marginal contribution to constrained inertia is tested.

Biplot scores (correlations between standardised environmental variables and standardised site scores) are extracted for triplot display.

2. **Identifying the recommended range.** The sensitivity plots are inspected for a stability region. A shaded band at 22 %–28 % (`shade_range=(0.22, 0.28)`) highlights the plateau where R_{adj}^2 remains strong before broader inclusion dilutes the reference set.
3. **F -critical indicator.** On the pseudo- F panel the F value at $p = 0.05$ and the threshold at which it is first reached are identified by linear interpolation.
4. **Selected threshold: 25 %.** `rda_threshold=0.25` places the decision near the centre of the 22 %–28 % stability band, balancing ecological signal strength with adequate sample size.

Outputs saved to `results/ref_threshold_sensitivity/`:

- **Tables:** per-score threshold metrics (`SumRel_threshold_metrics`, `MaxRel_threshold_metrics`), comparison summary (`threshold_comparison_summary`), RDA axes summary and terms summary at 25 % for each score.
- **Figures:** per-score sensitivity plots (2-panel: R^2 and F vs. threshold), threshold comparison overlay, RDA triplots at 25 % for each score.

5 Final Decisions

- **Scoring rule:** SumRel (cumulative contamination view).
- **Reference threshold:** 25 % of least-contaminated sites.
- **Final RDA at 25 %:** the integrated pipeline produces triplots, axes-summary tables, terms-summary tables, threshold-sensitivity curves, and comparison tables.

6 Code Organisation

All computation is implemented in the `src/zci` package:

`run_stage1.py`

Runs PCA + dual scoring (SumRel / MaxRel).

`run_stage_threshold.py`

Runs Stage 1, sweeps thresholds from 5% to 99% (step 2%), compares SumRel vs MaxRel, and exports sensitivity plots/tables plus final RDA outputs at 25%.

The standalone `run_stage_rda.py` is superseded by the integrated threshold-sensitivity pipeline, which performs the same RDA fitting at the selected threshold while also providing the full sensitivity analysis.